

07 • Interpretability

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All four tools hook into the **same** generation path as real turns, `sample`, and tool loops, and are tagged to conversation structure. They apply to **local ModelParticipants only** — an `APIParticipant` raises on any `interp` request rather than silently no-op'ing. Pass them to `conv.step(...)` or `conv.sample(...)`.

Activation capture

`conv.capture(...)` is a context manager: every `step/sample` inside the block records activations into a fresh `ActivationCache`, auto-tagged by speaker + turn.

```
with conv.capture(sites=["residual"], layers=[8, 12, 16],
                 offload="cpu") as cache:
    conv.step(conv.participant("bob"))
    conv.step(conv.participant("alice"))

# Query the cache by structure:
act = cache.at(participant="bob", layer=12, site="residual") #
# exactly one tensor, [seq, d_model]
records = cache.query(participant="bob") #
# all matching ActivationRecords
```

- **sites:** "residual" (per-layer residual stream), "attn" (attention sublayer output, post-o_proj), "mlp" (MLP output). Capture is a single clean forward over the full prompt+generation.

- **layers**: a list, or None for all layers. Keep it **narrow** — all layers × all tokens × many rollouts OOMs fast.
- **offload**: "cpu" moves tensors off-GPU as captured (essential for sweeps; uses a batched pinned transfer under the hood — ~7× faster than naive `.to('cpu')`); None keeps them on-device.

Each

ActivationRecord carries `.participant`, `.message_idx`, `.layer`, `.site`, `.tensor` (`[seq, d_model]`), `.token_span` (`prompt_len`, `seq_len`), and `.phases` — a `{prompt/reasoning/answer: (start, end)}` map so you can slice reasoning-vs-answer activations for CoT models:

```
rec = cache.query(participant="bob", layer=12)[0]
a0, a1 = rec.phases["answer"]
answer_acts = rec.tensor[a0:a1]          # activations over just the
                                         answer tokens
```

Build a probe direction from captured activations

```
import torch
# e.g. mean residual at layer 12 over the answer span for two
# conditions → a difference-of-means direction
pos = cache.at(participant="bob", layer=12)[rec.phases["answer"]
[0]:].mean(0)
# (collect a `neg` from another branch/condition similarly)
direction = (pos - neg)
direction = direction / direction.norm()
```

Steering — add or ablate a direction

SteeringSpec registers forward hooks that either **add** `coef * direction` to the residual at layers, or **ablate** (project out) that direction.

```
from interlens import SteeringSpec
steer = SteeringSpec(direction=direction, layers=(8, 12), coef=6.0,
mode="add")
msg = conv.sample("bob", "How do you feel about the proposal?",
steering=steer) # ephemeral, steered
```

```

ablate = SteeringSpec(direction=direction, layers=(12,),
                      mode="ablate") # remove the component
conv.step(conv.participant("bob"), steering=ablate)
# committed, ablated turn

```

`direction` is a `[d_model]` tensor on any device (moved to match). A summary (mode, layers, coef, direction norm) is recorded in `msg.metadata["steering"]` so a steered turn is reproducible. Steering **disables KV reuse** automatically (the intervention wasn't in the cached KV).

Activation patching — cross-branch causal tracing

Patch overwrites a decoder layer's residual at specific token positions with activations captured elsewhere (e.g. from another branch). This is the causal-tracing primitive: capture at turn N in one branch, inject at the aligned positions of another branch's forward.

```

from interlens import Patch
# 1) capture bob's layer-12 residual in the clean branch
clean = conv.branch()
with clean.capture(sites=["residual"], layers=[12]) as cache:
    clean.step(clean.participant("bob"))
donor = cache.at(participant="bob", layer=12) # [seq,
d_model]
# 2) inject those activations at chosen positions in a corrupted
branch's next forward
positions = (3, 4, 5)
patch = Patch(activations=donor[list(positions)], layer=12,
              positions=positions)
corrupt = conv.branch()
corrupt.transcript[-1] = ... # (corrupt
the setup somehow)
patched_msg = corrupt.sample("bob", patch=patch) # bob
generates with layer-12 residual patched

```

Aligning positions across branches is the caller's responsibility; Patch just performs the overwrite (only on the prefill forward, not per-token decode). Patching also disables KV reuse.

Token logprobs / surprisal / entropy

```
msg = conv.sample("alice", "State your final answer in one word.",
                  return_logprobs=True)
md = msg.metadata
md["logprobs"]    # per generated token: log p(token)
md["surprisal"]   # -logprob (nats)
md["entropy"]     # full next-token distribution entropy at each
                  # step (model uncertainty)
```

These are scalar-per-token lists (safe to store in metadata). Works with step too.

Next: [advanced pipelines that combine these](#), or [scaling to rollouts](#).